Aadhaar Friction Atlas

Measuring the Hidden Cost of Identity Maintenance

Theme: Unlocking Societal Trends in Aadhaar Enrolment & Updates

Focus: Citizen-centric analytics | Policy insights | Lifecycle analysis

‘High Aadhaar updates do not always indicate efficiency they often signal friction.’

UIDAI DATA HACKATHON

Team Id:- UIDAI_10578

Team Member:-

1. Sanika Deshmukh(Team leader)
2. Divya Addagatla
3. Pragati Kharat

Problem Statement

Aadhaar coverage is widely tracked through enrolment numbers, but the effort required to maintain Aadhaar records via frequent biometric and demographic updates is rarely quantified. Repeated updates impose a hidden administrative and citizen burden, especially on children, migrants, and residents of rapidly urbanizing regions.
Current analyses treat update volumes as operational success, overlooking system inefficiencies, poor initial capture quality, and access barriers.

Objective

To quantify Aadhaar maintenance burden and identify regions where identity upkeep outweighs enrolment expansion.

Datasets Used

- Aadhaar Enrolment Dataset
  - date, state, district
  - age_0_5, age_5_17, age_18_greater

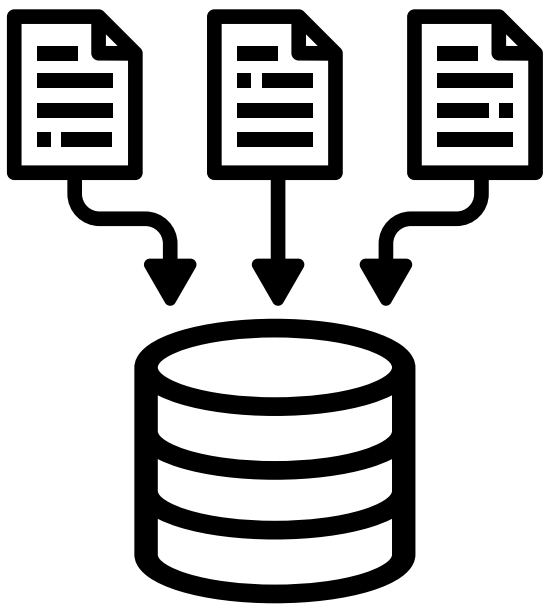
  Used as population denominator

- Aadhaar Biometric Update Dataset
  - bio_age_5_17, bio_age_17_

  Measures physical identity revalidation

- Aadhaar Demographic Update Dataset
  - demo_age_5_17, demo_age_17_

  Captures data instability & migration effects



Methodology

1. Combined multiple CSV files per dataset
2. Parsed dates → monthly buckets
3. Aggregated at district-month level
4. Handled missing values and divide by zero cases
5. Normalized all indicators using Min–Max scaling

Logic Behind the Code

Aadhaar Update Friction Index (AUFI)

AUFI is a composite score built from four normalized indicators:

1. Demographic Update Rate (DUR)– frequency of personal detail changes
2. Biometric Update Intensity(BUI) – repetition of biometric captures
3. Child Lifecycle Pressure(CLP) – higher updates due to growth in children
4. Maintenance Dominance Ratio (MDR)– updates compared to new enrolments

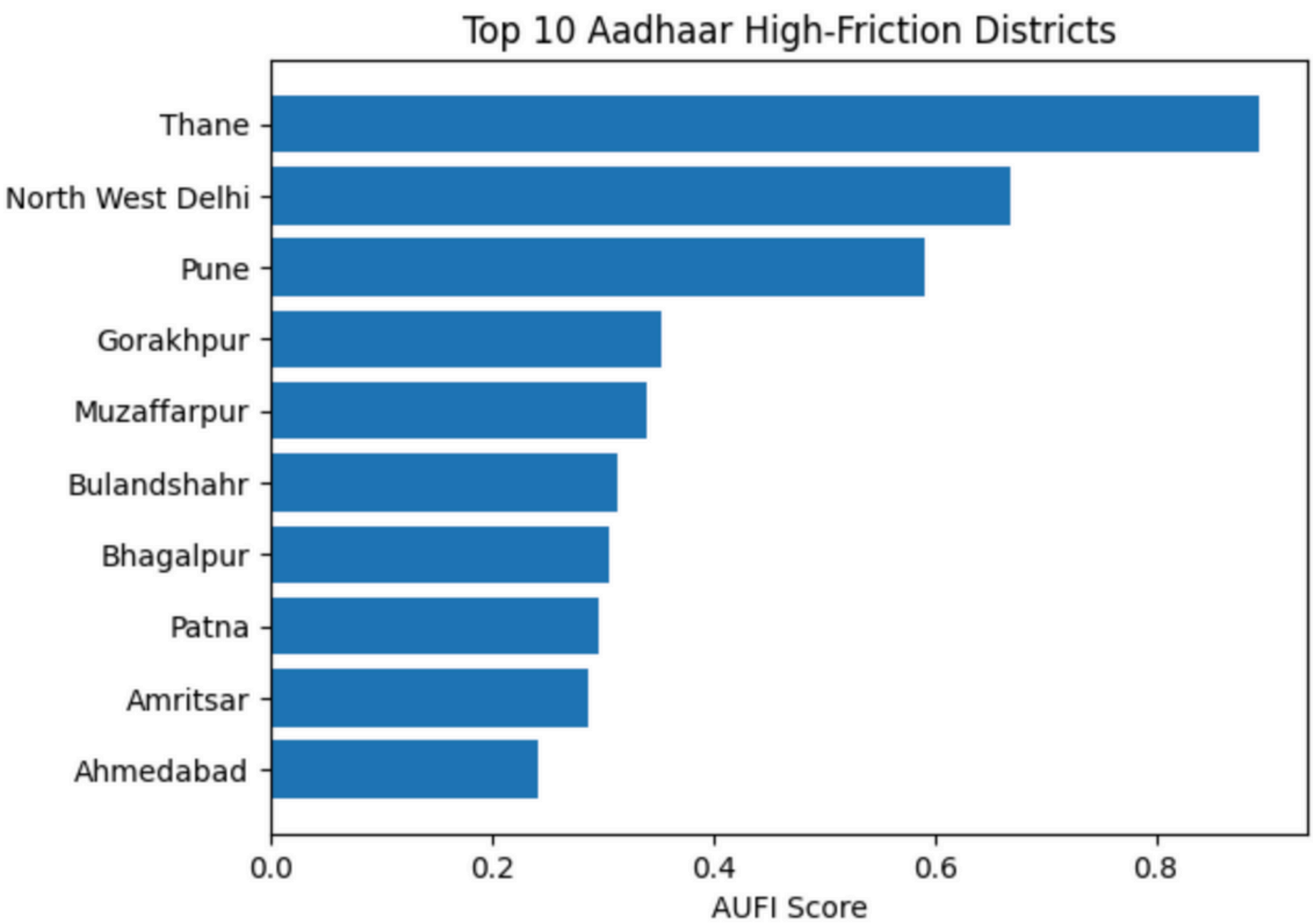
$$AUFI = \frac{DUR+BUI+CLP+MDR}{4}$$

ANALYSIS & INSIGHTS

High Friction ≠ High Enrolment :
Indicates maintenance-heavy Aadhaar systems.

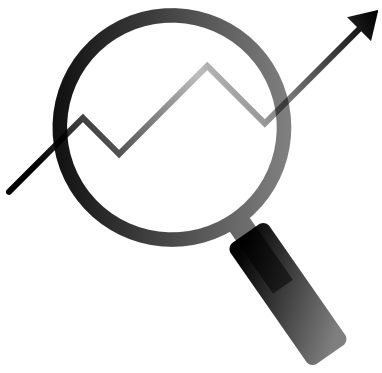
Child Lifecycle Burden is Significant :
Repeated re-validation in 5–17 age group.

Migration & Urban Pressure Hotspots :
Frequent updates in urbanizing districts



AUFI exposes districts where Aadhaar is hardest to maintain, not just most widely adopted.

Visual Insights



What the Visuals Reveal?

- Geographic clustering of high AUFI
- Certain regions consistently show elevated Aadhaar friction
- Weak correlation between enrolment and AUFI
- High enrolment does not necessarily imply higher friction
- Persistent regional patterns
- Friction hotspots remain stable across time periods

Code Snippet

CORE IDEA

```
# Compute AUFI core components
df['DUR'] = (df['total_demo'] / df['total_enrol']) * 1000
df['BUI'] = (df['total_bio'] / df['total_enrol']) * 1000
df['CLP'] = ((df['bio_5_17'] + df['demo_5_17']) / df['enrol_5_17']) * 1000
df['MDR'] = (df['total_demo'] + df['total_bio']) / df['total_enrol']

# Handle invalid values
df.replace([np.inf, -np.inf], 0, inplace=True)

# Normalize and compute AUFI
metrics = ['DUR','BUI','CLP','MDR']
df['AUFI'] = df[metrics].apply(
    lambda x: (x - x.min()) / (x.max() - x.min())
).mean(axis=1)
```

- For each district and month:
  - Calculate demographic updates per enrolment
    Calculate biometric updates per enrolment
  - Measure update pressure on children (5–17 age group)
  - Compare total updates with new enrolments
  - Normalize all indicators to a common scale
  - Average the indicators to compute AUFI score

Impact & Policy Recommendations

- Why AUFI Matters :
  - Identifies districts where Aadhaar is hardest to maintain.
  - Shifts focus from enrolment numbers to citizen efforts.
  - Enables data-driven planning of Aadhaar infrastructure.

- Policy Recommendations :
  - Deploy mobile Aadhaar units in high-AUFI districts.
  - Improve initial biometric capture to reduce repeat updates
  - Allocate staff dynamically based on AUFI trend

REDUCING AADHAAR FRICTION IMPROVES BOTH CITIZEN EXPERIENCE AND SYSTEM EFFICIENCY.

‘Aadhaar success should be measured not just by coverage, but by the effort required to maintain identity.’

Thank You!

Link for code